

M.Tech. (Online) Project Proposal and Work Plan

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Month and Year of Enrolment into the Degree: August, 2024

Stream: AI / ~~DSBA~~ / ~~ECE~~ (*Strike out whichever is not applicable*)

No. of Core Course Credits Completed: 16

No. of Elective Course Credits Completed: 20

1. Project Title

Detecting AI-Generated and Cloned Speech to Strengthen Voice-Based Authentication in Banking Environments

2. Problem Description

The proliferation of neural voice synthesis and cloning technologies has introduced a significant and underappreciated threat to voice-based authentication systems deployed in banking and financial services. Systems that once leveraged the uniqueness of human vocal characteristics as a security credential are now confronted with adversaries capable of generating convincingly realistic speech from just a few seconds of a target's voice.

This project focuses on building a reliable, automated pipeline to differentiate between genuine human speech and AI-generated or cloned audio. Framing this as a binary classification problem, the following research questions drive the investigation:

1. Can supervised machine learning classifiers reliably distinguish real human speech from AI-synthesised utterances, and what level of accuracy is achievable on unseen speakers?
2. What acoustic and signal-level properties - such as spectral smoothness, phase coherence, or prosodic naturalness - are most discriminative between cloned and authentic speech?
3. How do detection models hold up under conditions typical of banking telephony channels, including short utterance lengths, codec compression, and ambient noise?
4. Do deep learning architectures offer a meaningful performance advantage over classical machine learning approaches when evaluated on the same feature sets and data conditions?

The intended outcome is a validated classification framework that can be practically integrated into existing voice biometric pipelines used for fraud prevention and transaction authorisation.

3. Problem Motivation

Voice authentication has become a routine component of customer verification workflows in banks and call centres. Its appeal lies in convenience - customers can confirm transactions or prove their identity by simply speaking. However, this same convenience has become a liability: neural text-to-speech (TTS) and voice conversion systems, some available as open-source tools, can now clone a person's vocal identity with alarming fidelity.

The downstream consequences of a successful voice spoofing attack are concrete and severe. Unauthorised fund transfers, fraudulent identity verifications, and social engineering attacks facilitated through synthesised

voices represent direct financial and reputational risks to institutions and customers alike. Existing voice biometric systems were largely designed against replay attacks or basic synthesis methods - not against the diffusion model or transformer-based neural TTS systems that are state-of-the-art today.

There is therefore a pressing need to develop detection mechanisms that are not only accurate in lab conditions but also robust to the degraded audio quality typical of real banking calls. A well-validated detection system would serve multiple purposes:

- Hardening existing voice biometric authentication against spoofing
- Providing an additional fraud detection signal in call centre pipelines
- Supporting forensic audio analysis in dispute resolution
- Helping institutions meet emerging regulatory expectations around AI-generated media

Beyond the banking sector, the core research has relevance to media authentication, misinformation forensics, and digital identity protection more broadly.

4. Related Work and Novelty

The detection of synthetic speech - commonly framed as anti-spoofing within the automatic speaker verification (ASV) community - is a reasonably well-studied area. The ASVspoof challenge series (2015 through 2021) has served as the primary benchmarking platform, evaluating systems against logical access (TTS, voice conversion) and physical access (replay) attacks using features such as MFCCs, Constant-Q Cepstral Coefficients (CQCCs), and spectrogram-based convolutional neural networks.¹

Deep learning architectures have progressively displaced classical approaches. CNNs, ResNets, and hybrid CNN-LSTM models have all demonstrated strong performance on ASVspoof datasets.² More recently, self-supervised representations such as those produced by wav2vec 2.0 have shown competitive results when fine-tuned for spoof detection, suggesting that large pretrained models carry acoustic priors useful for distinguishing real from synthesised speech.³

Despite this progress, most published work evaluates against benchmark datasets under controlled recording conditions. There is comparatively little work that directly addresses the conditions encountered in live banking telephony - particularly short transactional utterances (often just three to five words), audio passed through G.711 or similar codecs, and varying levels of environmental noise. This project addresses that gap specifically. The novelty contributions include:

- Constructing an evaluation set that explicitly simulates banking-channel conditions rather than clean studio recordings
- Performing a principled comparison between traditional ML baselines (SVM, Random Forest, Gradient Boosting) and deep learning models under identical data conditions
- Assessing model degradation curves as a function of SNR and compression level, which is rarely reported in anti-spoofing literature
- Applying feature interpretability tools (SHAP for classical models, gradient saliency for CNNs) to understand which signal-level properties most reliably betray synthetic origin

Footnotes:

1. Nautsch et al., ASVspoof 2019: Future Horizons in Spoofed and Fake Speech Detection (Odyssey 2020)
2. Todisco et al., ASVspoof 2019: Logical Access (INTERSPEECH 2019)
3. Baevski et al., wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations (NeurIPS 2020)

5. Proposed Solution Approach

The project follows a five-stage experimental pipeline: dataset construction, feature engineering, model development, robustness evaluation, and interpretability analysis.

Stage 1: Dataset Construction

The dataset will draw on publicly available genuine speech corpora - specifically LibriSpeech and VoxCeleb - supplemented by AI-generated speech produced using open-source TTS systems (e.g., Tacotron 2 with WaveGlow, VITS, or YourTTS). To accurately represent banking authentication scenarios, the dataset will be shaped as follows:

- Utterances will be trimmed or synthesised to fall within the 3–8 second range, mimicking typical transaction confirmation phrases
- Telephony conditions will be simulated by downsampling to 8 kHz and applying G.711 mu-law encoding
- White and babble noise will be added at SNR levels of 5, 10, 15, and 20 dB

All splits will enforce strict speaker independence between training, validation, and test sets to prevent identity leakage.

Stage 2: Feature Engineering

Two broad classes of features will be explored.

1. First, handcrafted acoustic features: MFCCs (13–40 coefficients with deltas), spectral centroid, bandwidth, roll-off, zero-crossing rate, pitch, harmonic-to-noise ratio, and phase spectrum statistics.
2. Second, time-frequency representations: log-Mel spectrograms and short-time Fourier transform (STFT) spectrograms suitable for input to image-based classifiers.

Stage 3: Model Development

Baseline models - SVM with an RBF kernel, Random Forest, and Gradient Boosting - will be trained on the handcrafted feature vectors.

Deep learning models will include a CNN classifier operating on log-Mel spectrograms and a CNN-LSTM hybrid that captures both local spectral patterns and temporal structure.

As a stretch goal, fine-tuning a pretrained wav2vec 2.0 encoder will be explored. Hyperparameter tuning will use stratified k-fold cross-validation.

Stage 4: Robustness Evaluation

Each model will be evaluated not only on clean audio but across the noise and compression conditions built into the dataset.

Performance will be reported as curves plotted against SNR and codec type, providing a practical view of deployment feasibility rather than just peak benchmark numbers.

Stage 5: Interpretability

SHAP values will be computed for the classical ML models to quantify feature contribution.

Gradient-based saliency maps will be generated for the CNN models to identify which time-frequency regions drive classification decisions.

The goal is to produce signal-level insights - ideally identifying consistent artefacts that betray synthetic generation - rather than treating the models as black boxes.

6. Evaluation Plan

Model performance will be quantified using **accuracy, precision, recall, F1-score, ROC-AUC, and Equal Error Rate (EER)** - the last being the standard metric in anti-spoofing literature, as it balances false acceptance and false rejection at a single operating point.

Experimental comparisons will be structured along four axes:

1. classical ML versus deep learning architectures;
2. clean audio versus noise and compression-degraded conditions;
3. long utterances versus short banking-style phrases; and
4. seen versus unseen TTS systems.

The final test set will deliberately include synthetic voices generated by systems not present during training, providing a measure of generalisation beyond the specific cloning tools used to build the dataset.

Results will be reported with **confusion matrices, ROC curves, and the degradation curves** described in Section 5.

7. Feasibility, Hardware, and Risk Assessment

The project is well within the scope of completion during the planned timeline. All required datasets are publicly available and the TTS systems needed to generate synthetic samples are open-source. The primary software stack - **Python, Librosa, PyTorch, Scikit-learn, and SHAP** - is freely accessible.

Hardware Considerations

Training will be run on two platforms.

1. Google Colab provides access to T4 or A100 GPUs sufficient for all CNN and CNN-LSTM experiments.
2. The personal RTX 3060 (6 GB VRAM) laptop is well-suited for local development, feature extraction, and classical ML training.

For context:

- The CNN models targeted in this project - operating on 128-bin log-Mel spectrograms with inputs under 10 seconds - have a parameter count in the low millions, comfortably fitting in 6 GB of VRAM with batch sizes of 32–64.
- The CNN-LSTM hybrid may require smaller batches on local hardware but will not require a high-end workstation.
- Fine-tuning wav2vec 2.0 (the most compute-intensive step) is best done on Colab. A practical split would be: run exploratory experiments and debugging locally on the RTX 3060, and schedule longer training runs overnight on Colab.

Known Risks and Mitigations

- Diversity of synthetic samples: Using multiple TTS systems (Tacotron 2, VITS, YourTTS) reduces dependence on any single generation method
- Overfitting to seen TTS systems: Strict speaker- and system-independent test sets, combined with data augmentation, will stress-test generalisation
- Compute constraints: Model complexity will be managed adaptively; if resources are insufficient for full wav2vec fine-tuning, feature extraction from a frozen pretrained encoder is a viable fallback
- Dataset size: LibriSpeech and VoxCeleb together provide tens of thousands of utterances; combined with augmented synthetic samples, class balance is achievable

8. Work Plan

Phase 1 - Term: January 2026

Task	Deliverables	From	To
Literature review and benchmark survey	Written survey document summarising key anti-spoofing methods and datasets	Jan 2026	Feb 2026
Dataset collection and preprocessing	Curated, balanced dataset with genuine and synthetic utterances	Feb 2026	Mar 2026
Feature extraction pipeline development	Modular Python preprocessing scripts with logging and reproducibility	Mar 2026	Apr 2026

Phase 2 - Term: Summer 2026

Task	Deliverables	From	To
Baseline ML model training and tuning	Trained SVM, Random Forest, and Gradient Boosting classifiers with cross-validation results	May 2026	Jun 2026
Deep learning model development (CNN, CNN-LSTM)	Trained and validated deep models; comparative performance report	Jun 2026	Jul 2026
Robustness experiments across noise and compression conditions	Degradation curves and SNR sensitivity analysis	Jul 2026	Jul 2026

Phase 3 - Term: August 2026

Task	Deliverables	From	To
Interpretability analysis (SHAP, saliency maps)	Feature importance report identifying key discriminative signal characteristics	Aug 2026	Sep 2026
Final evaluation and statistical testing	Complete performance metrics, ROC curves, EER comparisons	Sep 2026	Oct 2026
Thesis writing and submission	Final project report in institute-prescribed format	Oct 2026	Nov 2026

We approve the project proposal and workplan given above.

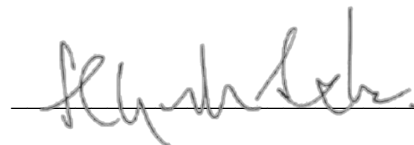


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